

Computer Systems Fundamentals

Mock Test — Questions & Answers Reference Sheet

TEST 1 — Data Representation & Number Systems

Q1 [10/10] — If you have 500 images each of 1024×768 pixels, what is the total disk space (in MB) that these images will occupy (assume 24-bit RGB format)?

✓ 1125 MB (or 1,125)

Calculation: $500 \times 1024 \times 768 \times 3 \text{ bytes} = 1,179,648,000 \text{ bytes} \div 1,048,576 \approx 1125 \text{ MB}$

Q2 [10/10] — Convert the Denary value 75 into its 8-Bit Binary equivalent, then convert that Binary value into Hexadecimal.

✓ 8-Bit Binary: **01001011**

✓ Hexadecimal: **4B**

Q3 [5/5] — What is the hexadecimal representation of the binary 11010011?

✓ D3

Split: $1101 = D$, $0011 = 3 \rightarrow D3$

Q5 [5/5] — What is the denary equivalent of the hexadecimal B3?

✓ 179

$B = 11$, $3 = 3 \rightarrow (11 \times 16) + 3 = 176 + 3 = 179$

Q6 [15/15] — Convert the IEEE754 Hexadecimal representation 420A0000 to Decimal.

✓ 34.5 (or +34.5)

Q7 [0/5 ■ Incorrect in test] — What is the binary representation of the hexadecimal 4C?

✓ 01001100

$4 = 0100$, $C = 1100 \rightarrow 01001100$ (incorrect answer given during test was 01111100)

Q8 [10/10] — What is the result in denary of performing this binary subtraction? $10011001 - 01100110$

✓ 51

$10011001 = 153$, $01100110 = 102$, $153 - 102 = 51$

Q9 [2.5/2.5] — If you perform an AND operation on the binary numbers shown below, what would the answer (in binary) be? 10101011 AND 00110011

✓ 00100011

AND each bit column: only bits where BOTH are 1 remain 1

Q10 [5/5] — What is the largest unsigned denary value that can be stored in a 16-bit system?

✓ 65535

$2^{16} - 1 = 65535$

Q11 [10/10] — What is the 8-Bit Binary Two's Complement representation of the signed Integer -108?

✓ 10010100

108 = 01101100, invert = 10010011, add 1 = 10010100

Q12 [5/5] — What is the denary equivalent of the binary 11011110?

✓ 222

128+64+16+8+4+2 = 222

Q13 [15/15] — Convert the Decimal number +27.75 into IEEE754 format.

✓ 41DE0000 (hex)

Sign=0, exponent=10000011 (131), mantissa=10111100000...

TEST 2 — Computer Systems & Networks

Q1 [8/8] — The CPU can only perform simple arithmetic. Tick ALL correct operations the CPU can perform directly.

- ✓ Multiply a number by 2
 - ✓ Subtract one number from another number
 - ✓ Divide a number by 2
 - ✓ Add one number to another number
 - ✗ Multiply one number by another number
 - ✗ Multiply a number by 3
 - ✗ Divide a number by 3
 - ✗ Divide one number by another number
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Q2 [5/5] — Which of the following statements are correct? (Tick all correct statements)

- ✓ UTP stands for Unshielded Twisted Pair
 - ✓ LAN stands for Local Area Network
 - ✓ STP stands for Shielded Twisted Pair
 - ✓ MAN stands for Metropolitan Area Network
 - ✓ SAN stands for Storage Area Network
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Q3 [10/10] — What is the first valid host address for the network which includes the IP address 163.56.170.94/22?

✓ 163.56.168.1

Network address: 163.56.168.0 → first host: .168.1

Q4 [5/5] — In a round robin process management system, each process is able to access the CPU as and when needed. Is this true or false?

✓ False

Round robin allocates fixed time slices; processes do NOT access the CPU on demand.

Q5 [5/5] — In the context of this module, a quantum is a special branch of computing. Is this true or false?

✓ False

In this context, a 'quantum' refers to the fixed time slice used in round-robin scheduling, not quantum computing.

Q6 [16/16] — Given the Partition Boot Sector image (Partition_Boot_Sector_1.PDF / 4COSC004W-Test2-PSZdd_Image1.pdf), answer the following questions about the FAT16 partition.

- **Partition starts at sector:** 2048 (also accept 1048576)
 - **Size of Reserved Area:** 2 sectors
 - **Number of FATs:** 2
 - **Size of each FAT:** 7 sectors
 - **FAT start address (sector):** 2048 (also accept 1048576)
 - **Size of Reserved Area (for Root Dir calc):** 2 (also accept 2050 or 1049600)
 - **Total size of all FATs:** 14 sectors
 - **Start address of Root Directory:** 2064 (also accept 1056768)
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Q7 [5/5] — Given an IP address of 10.34.23.134 and a subnet mask of 255.255.0.0, the result of performing an AND operation on these two sets of numbers is 10.34.0.0. What is this known as?

✓ Network Address

Q8 [10/10] — A new process is about to be instantiated, which will require 125 kB of memory. The Operating System uses the Best-fit method for memory allocation (see schematic in test paper).

✓ Best-fit selects the smallest available memory partition that is still large enough to satisfy the request (125 kB).

Exact partition answer depends on the memory schematic shown in the test paper.

Q10 [20/20] — Given the disk sector zero image (43615_DSZ(1).pdf / 43615_DSZ.pdf), answer the following questions.

- **Number of partitions in this disk:** 4
 - **File system of 1st partition:** FAT 16 (also accept 04)
 - **Starting LBA address (little endian):** 02 02 00 00
 - **Converted to big endian:** 00 00 02 02
 - **Big endian value in denary:** 514 Sectors
 - **Size of 1st partition (little endian):** 00 C8 00 00
 - **Size in denary:** 51200 Sectors
 - **Partition size in KB:** 25600 KB
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